

Education

- 2015-2020 **Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering**
Indian Institute of Technology Madras, Chennai, India CGPA: 8.78
- 2015 **XII - Karnataka Board, KLE Society's Independent PU College, Bangalore** 97.30 %
- 2013 **X - ICSE, B P Indian Public School, Bangalore** 96.33%

Professional Experience

- Software Engineer, Level IV, Google LLC, New York**
- Dec 2022 - Present **Software Engineer, Level IV, Google India Pvt Ltd, Bangalore**
- Aug 2020 - Nov 2022 { Developing intelligent features for the Google Workspace Editors (Docs, Slides, Keep, etc) using my expertise on the products' client-side software, supporting tools and libraries, and natural language processing infrastructure.
 { Using cutting-edge frontend tools like Web Assembly and Emscripten, and Google-internal technologies like j2Cl, client-side cross-platform frameworks and build systems, to develop user-facing features such as spellcheck in encrypted documents for five languages and writing style suggestions for English text.
 { Formulating technical designs for independent end-to-end problems, driving cross-team collaboration, upholding software reliability practices, technical-debt resolution and documentation, and proactively identifying areas of future work.
 { Guiding junior engineers on programming and software design tasks to enable timely delivery of products to customers.
- May 2019 - July 2019 **Software Engineering Intern, Google India Pvt Ltd, Bangalore**
- Worked on the Editors client-side software infrastructure to develop a user interface with control options to undo or provide feedback on the correction and a logging framework, for the Google Docs text auto-correction feature.
- May 2018 - July 2018 **Research Intern, Big Data Experience Labs, Adobe Research, Bangalore**
- Developed a mobile application for Text to Scene Conversion in Augmented Reality, based on novel research techniques for prediction of three-dimensional object sizes and positions from textual features.

Research Experience

- Sep 2019 - May 2020 **Paraphrase Generation with a Bilingual Model and Continuous Embeddings** *Master's Thesis, Language Technologies Institute, Carnegie Mellon University*
- Machinated a novel technique for paraphrase generation using the [von Mises-Fisher \(vMF\) Loss](#) on a transformer network, and showed that it produces superior paraphrases as compared to the log-likelihood model by employing bilingual data to induce zero-shot paraphrasing, guided by [Prof. Yulia Tsvetkov](#).
- May 2017 - July 2017 **Cognitive Approach to Natural Language Processing**
- Research Intern, Department of Computer Science and Automation, Indian Institute of Science (IISc), Bangalore*
- Developed a cognitive text parser that combines syntactic and semantic approaches, to process textual data into cognitive structural representations, to be used as a feature extractor for downstream NLP tasks, and demonstrated the correlation of the extracted cognitive features with semantic and syntactic text features, guided by [Prof. Veni Madhavan](#).

Publications and Patents

- [Publication and Poster] **Improving the Diversity of Unsupervised Paraphrasing with Embedding Outputs (Paper, Poster)**
Monisha Jegadeesan, Sachin Kumar, John Wieting, Yulia Tsvetkov
 In [Workshop on Multilingual Representation Learning](#),
 The 2021 Conference on Empirical Methods in Natural Language Processing ([EMNLP 2021](#))
- [Publication and Poster] **Adversarial Demotion of Gender Bias in Natural Language Generation (Paper, Poster)**
Monisha Jegadeesan
 In [ACM CODS-COMAD 2020](#) - Young Researchers' Symposium
- [Poster] **ARComposer: Authoring Augmented Reality Experiences through Text (Poster)**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 In ACM User Interface Software and Technology Symposium 2019 ([ACM UIST 2019](#))
- [Filed Patent] **Visualizing Natural Language through 3D Scenes in Augmented Reality**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 Filed at the US PTO (Application Number: 16/247,235)

Projects

Real-Time AI-Powered Customer Support Chatbot (2024)

Designed and developed a multimodal, AI-powered chatbot to enhance customer support by leveraging vector databases and retrieval-augmented generation techniques. Utilized synthetic data generation and model-supported data creation to train the chatbot on company knowledge bases, significantly improving response times and reducing human agent workload by 35%. Collaborated with cross-functional teams to integrate language models and optimize data collection workflows, ensuring high-quality, scalable AI service delivery.

Tech: LangChain, OpenAI API, ChromaDB, Python

Customer Sentiment Analysis with NLP and Data Annotation (2024)

Engineered a comprehensive sentiment classification system by designing data markup and annotation pipelines that processed large-scale e-commerce customer reviews. Applied advanced natural language processing techniques including tokenization, TF-IDF vectorization, and logistic regression modeling to extract actionable insights. Enabled continuous quality evaluation and iterative refinement of AI models through model-supported data generation and human-in-the-loop data collection, achieving an 89% classification accuracy and identifying key improvement areas.

Tech: Python, NLTK, TF-IDF, Logistic Regression

Employee Attrition Predictive Analytics and Dashboard (2024)

Led the design and implementation of complex data collection and analysis workflows to model employee attrition using demographic and HR data. Coordinated data annotation efforts and quality assurance processes in collaboration with stakeholders, using scripting and SQL to streamline data processing. Developed predictive models with logistic regression and created interactive dashboards for monitoring attrition trends, directly enabling policy changes that reduced attrition rates by 12%.

Tech: SQL, Power BI, Python

Predictive Maintenance AI Model for Industrial IoT (2024)

Developed a predictive maintenance model leveraging time-series sensor data collected through both human-curated and synthetic data generation methods. Implemented feature extraction pipelines from multimodal IoT sensor inputs such as temperature, vibration, and pressure data. Employed Random Forest classifiers, with ongoing model evaluation supported by scripted data analysis tools to enhance AI model accuracy and operational performance. The solution reduced unexpected machine downtime by 28% and optimized maintenance schedules.

Tech: Python, Random Forest, Sensor Data Processing

About :

I am focused on advancing AI capabilities through the development and evaluation of complex, multimodal datasets, utilizing synthetic data generation, model-supported data creation, and human-in-the-loop approaches. Experienced in designing and coordinating data collection efforts that meet rigorous quality standards, analyzing large-scale language and multimodal data, and building tools using Python for data analysis and creation. Skilled in collaborating with cross-functional teams including data scientists, engineers, and product managers to drive innovation in AI model training and evaluation. Proficient in computational linguistics, AI data creation, and leveraging state-of-the-art methods to bootstrap and test new AI functionalities. Committed to delivering best-in-class AI data services in fast-paced, collaborative environments.

